CivicPulse Gauteng

- Deprivation, political geography & voter participation

- Turning data into insights, and insights into better decisions.

Presented by

University of South Africa

DIRISA SDC 2026 · TEAMS QUALIFICATION SUBMISSION

CivicPulse Gauteng

Why do some wards vote, and others don't?

A complete ward-level analysis on IEC 2021 IEC 2016 merged (242 wards matched); ward Census still unlocks full H1

Johannesburg and Tshwane, ahead of the 4 November 2026 elections.

242
wards analysed

195,424
ballot rows cleaned

6.3pp
model error

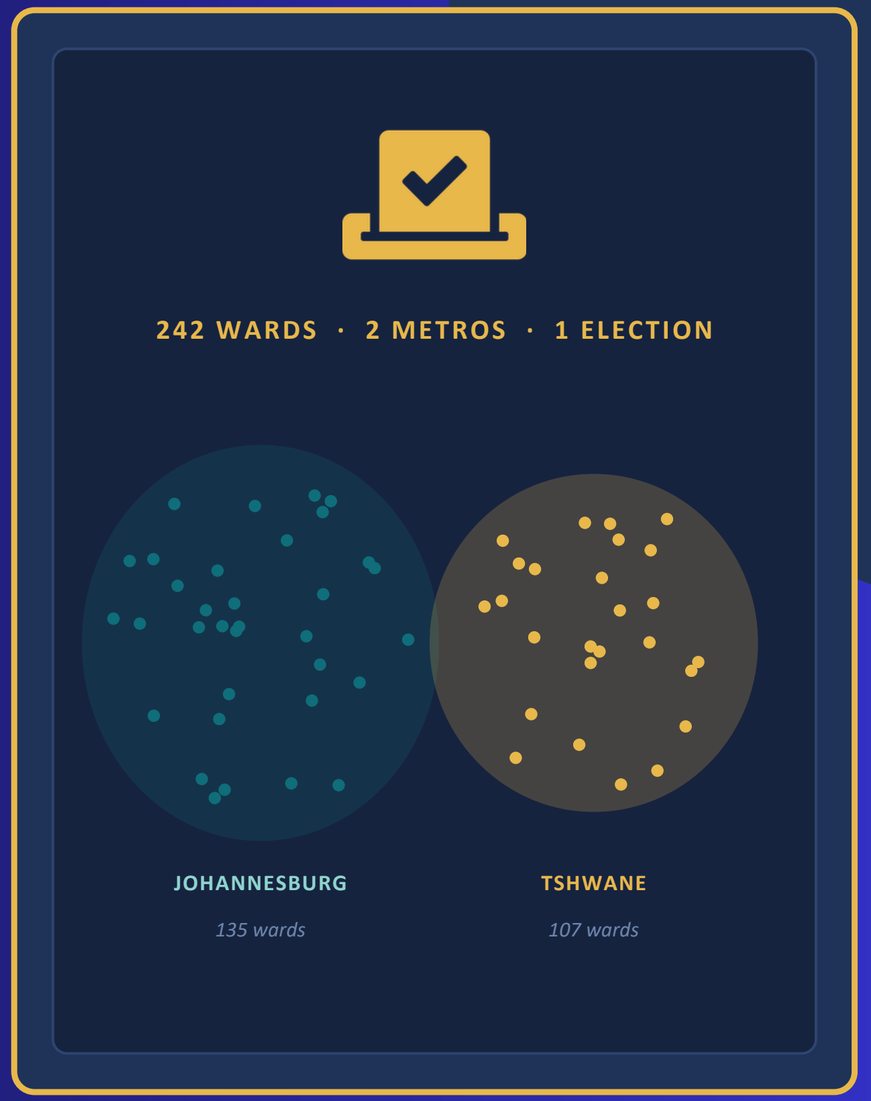


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01 · THE QUESTION

One question, two testable claims

To what extent do multi-dimensional deprivation and ward-level party dominance explain differences in voter turnout especially among young voters across Johannesburg and Tshwane?

H1

Deprivation friction

Wards with worse living conditions, poor water access, informal dwellings, irregular refuse collection turn out to vote less.

H2

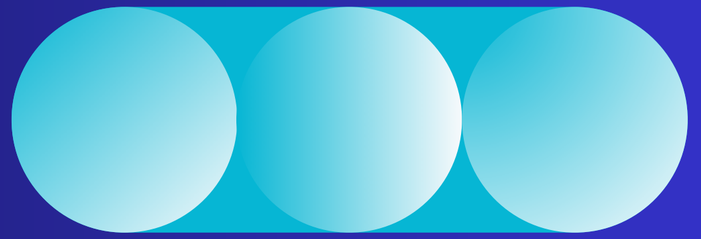
Incumbency demobilisation

Where one party has held a ward for years, do voters stay home instead of switching parties? We test that and let the data disagree if it must.

Unit of analysis: 242 wards (135 Johannesburg, 107 Tshwane) · A negative result still counts as a finding

01 · OBJECTIVES

What we set out to do



Like the protest-hotspot teams, we use public data early but for elections. Four aims before the November 2026 local government elections:

1 Map the metros

Build a ward-level turnout picture for Johannesburg and Tshwane so IEC and NGOs know where to focus.

2 Test the drivers

Check whether service delivery stress and long one-party control line up with lower participation.

3 Earn trust

Outreach grounded in ward evidence — not generic youth-blame campaigns.

4 Respect dignity

Participation is a civic right; poor services should not mean being written off.

242 wards · 135 Johannesburg · 107 Tshwane

01 · AUDIENCES

Who benefits from CivicPulse

1	**IEC & civil society**	Where to aim registration and turnout drives with evidence people can trust
2	**Political parties**	Whether safe seats are quietly demobilising their own voters
3	**Journalists**	Evidence that goes beyond the well-worn “youth apathy” narrative
4	**Municipalities**	Whether service delivery failures show up in civic participation citizens not written off

03 · CLEANING & PROCESSING

Before any model runs, the data has to earn trust

8/8

Census files found at municipality level, not ward caught by an audit before it broke anything

1

Duplicate file caught by MD5 hash: a Johannesburg age file re-uploaded under a new name

45,086

of 195,424 ballot rows kept presidential ballots dropped, since they're separate votes by the same people

5

voting districts dropped where votes exceeded the roll impossible turnout, listed in the notebook

03 · FEATURE ENGINEERING

Every feature exists because a hypothesis needs it

Turnout & fragmentation

Ward and VD turnout; effective number of parties (ENP)

Representation

Winner, winner share, margin, party shares (ANC/DA/EFF/ActionSA)

Deprivation (H1)

Water, dwelling, refuse shares → one standardised index

Change over time

2016→2021 retention, Δturnout, Δincumbent share (H2)

Youth target (T2)

Census cohort aged 14–25 in 2022, aged to the 2026 electorate

Geographic proxy

Tent voting stations as a sub-ward informal-settlement marker

```
turnout = (valid + spoilt) / registered
```

$$ENP = 1 / \Sigma\ s_i^2$$

Turnout splits by party, not by metro

42.8%

mean ward turnout

range 21.1%–65.7%, near-symmetric

39.5% vs 51.7%

median turnout, ANC- vs DA-won wards

the clearest split in the data

42.0% vs 44.6%

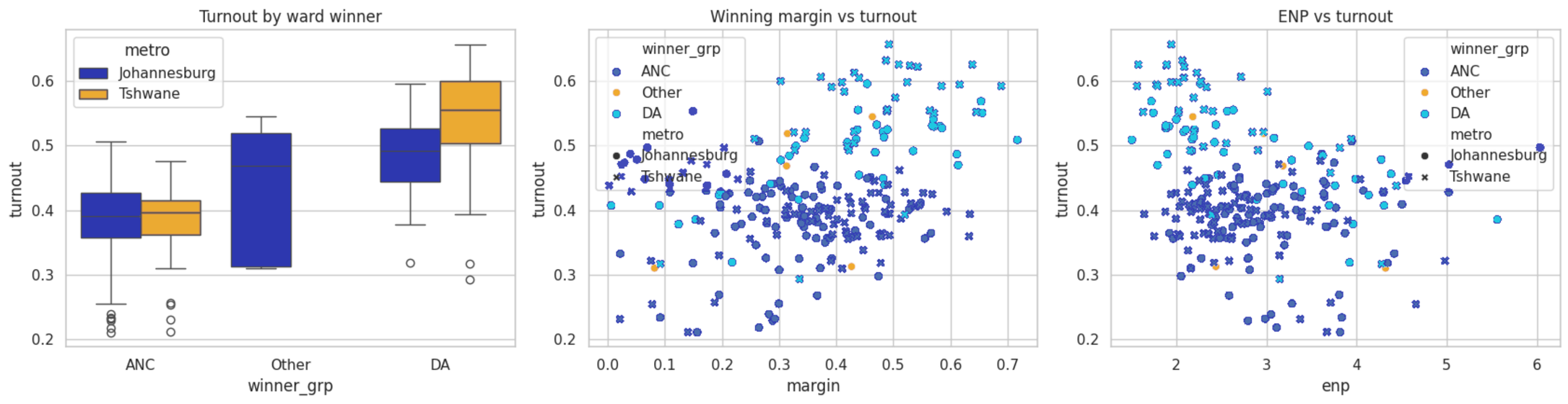
JHB vs Tshwane turnout

not the main story (−14.6pp mean turnout 2016→2021)

ρ = −0.78

EFF vote share vs turnout

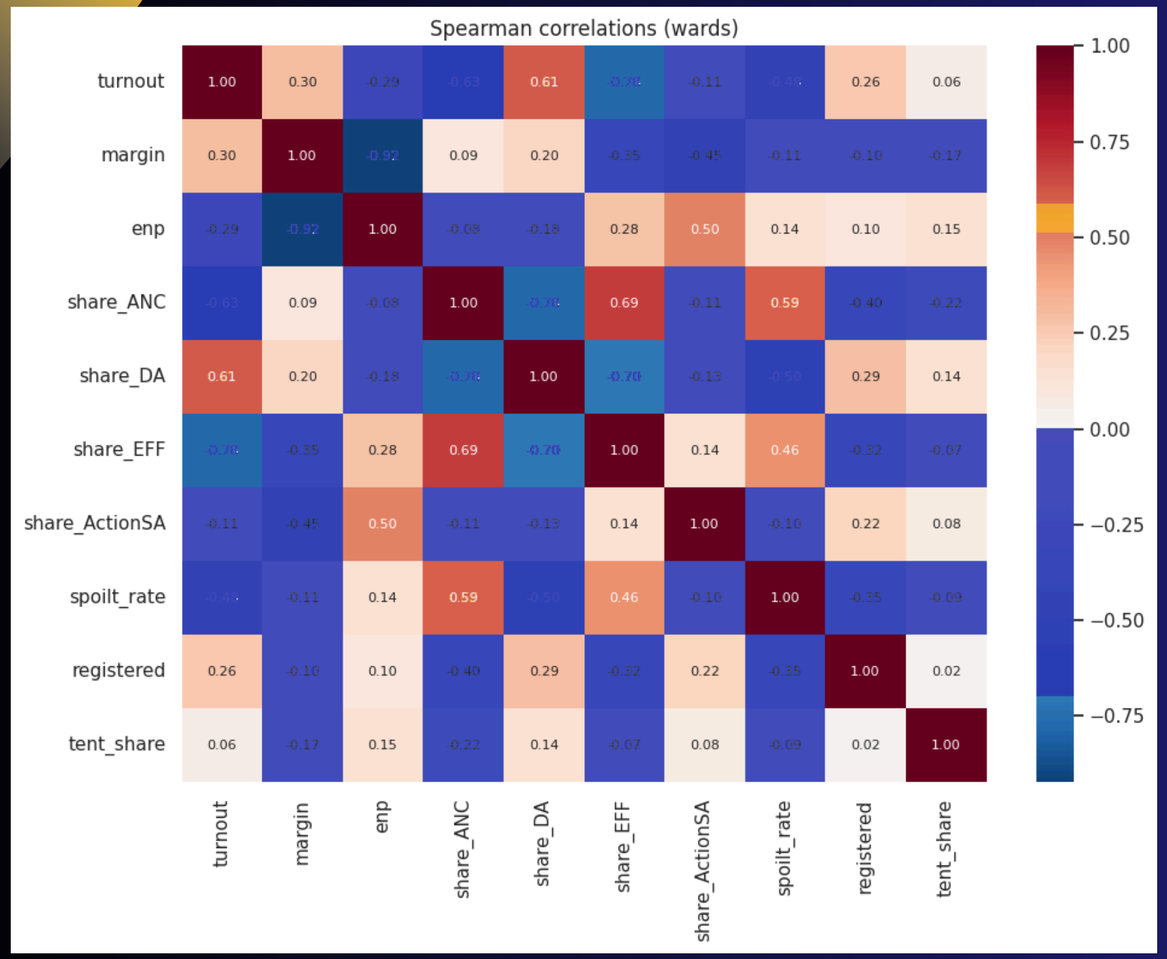
DA share correlates +0.61



From CivicPulse_Submission.ipynb · turnout by ward winner; winning margin vs turnout; ENP vs turnout (242 wards)

What moves together with turnout

Spearman correlations across 242 wards from CivicPulse_Submission.ipynb §5.4; p-values FDR-corrected in the notebook table



KEY READS FOR THE PITCH

- turnout ↔ share_ANC: ρ ≈ −0.63
- turnout ↔ share_DA: ρ ≈ +0.61
- margin ↔ ENP: ρ ≈ −0.92
- tent_share ↔ turnout: weak (H1 proxy inconclusive)
- 8 of 9 feature pairs significant at q < 0.05 after FDR

Engineered panel heatmap (turnout_change, deprivation, youth): figures/eda/06_correlation_heatmap.png in civicpulse_gauteng

05 · STATISTICAL TESTING

Every test earns its place

Chosen for small n, skewed features, and repeated comparisons.

Test	Chosen because	What it showed
Kruskal–Wallis by winner party	Non-normal turnout, 3+ groups	$H = 107.5$, $p = 4.6 \times 10^{-24}$ — party is the major split
Spearman + FDR correction	Robust to skew, controls false discoveries	8 of 9 correlations significant at $q < 0.05$
Interaction model (M3)	Tests whether dominance cuts both ways	DA wards: +0.41 SD, p<.001 — ANC wards: +0.05 SD, p=.58
Within-ward fixed effects	Holds ward politics constant for the tent proxy	Tent VDs −1.0pp, 95% CI [−3.1, +1.1] — not significant

Caution: these are cross-sectional associations they show correlation within one election, not proof of cause.

Ridge regression, picked by a rule set before results came in

Model	MAE (pp)	R^2
Baseline (mean)	8.57	0.00
OLS	6.33	0.365
Ridge — chosen	**6.33**	**0.365**
Random Forest	6.28	0.373
Gradient Boosting	6.55	0.308
XGBoost	6.50	0.348

One-standard-error rule

Random Forest wins narrowly (6.28 ± 0.27), but Ridge falls inside that margin so the simpler model is chosen.

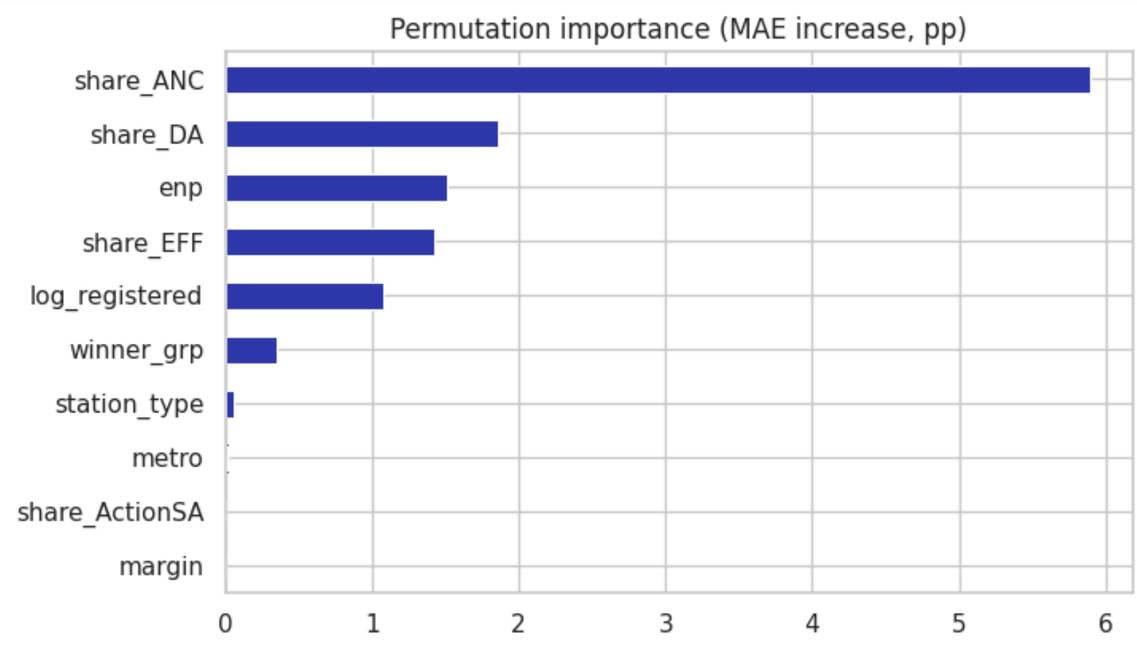
Regression, not just classification

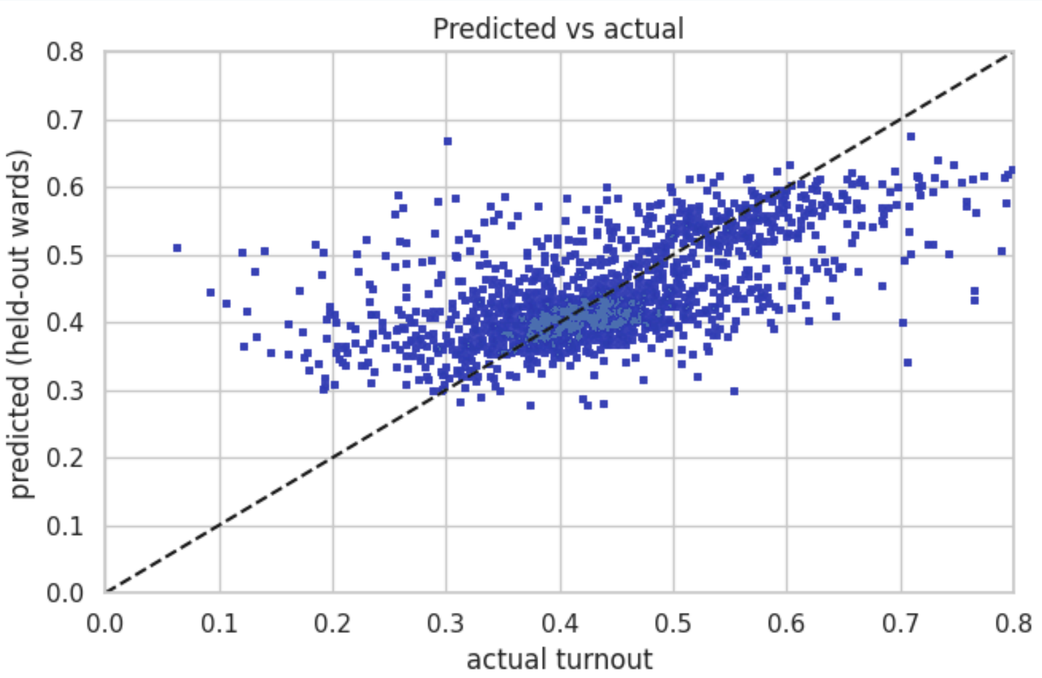
Ranking Ridge's predictions gives AUC 0.76, matching dedicated classifiers one model serves both jobs.

6.3 points of error

Against a 44-point spread between the lowest and highest wards, that's a usable signal.

What the model actually leans on





Permutation importance: share_ANC dominates (~5.9pp MAE increase when shuffled), followed by share_DA, ENP, and share_EFF. **Predicted vs actual:** held-out wards cluster near the diagonal, with more spread at the extremes consistent with a 6.3pp average error.

06 · MODELLING

What the 2016 merge actually unlocks

2026 FORECAST MODEL

2.44pp

held-out MAE forecasting 2021 turnout from 2016 (lagged) alone vs 6.32pp for the cross-sectional model

R^2 = 0.876 (vs 0.365 cross-sectional). Turnout is highly persistent knowing a ward's 2016 result predicts 2021 far better than political geography alone.

Real example: Ward 79900081 (Tshwane):

`2021 turnout 21.2%  →  2026 forecast 11.9%  (90% interval 7.2%–16.4%)`

MOBILISATION GAP

−44.9pp

the largest gap in the dataset: actual turnout minus what the model expects from a comparable place

Chris Hani Flats voting station (Tshwane): 6.2% actual vs 51.2% predicted turnout comparable places vote 8× more.

Real example: Ward 79900059 (Tshwane, DA):

`turnout 29.4%, ward-level gap −17.2pp — the largest ward-level shortfall`

This is exactly what the Priority List view in the deployed app ranks by not low turnout alone, but turnout that's low relative to comparable places.

Reading the outputs back against the original question

H1

Deprivation friction

INCONCLUSIVE

Tent proxy: −1.0pp within wards, not significant.

What would change it:

Needs ward-level Census water/dwelling/refuse shares still only District-level on re-export.

H2

Incumbency demobilisation

NOT SUPPORTED — TESTED TWICE

Cross-section: DA strongholds turn out; ANC wards don't. Over time (2016→2021, real data): retained wards fell 14.6pp, flipped wards 15.5pp not significantly different (p=.96).

The mechanism, now tested:

In retained wards, turnout fell 14.6pp vs the incumbent's vote share falling 13.1pp disengagement outweighs defection, but holding the ward isn't what drives it.

T2

Youth registration gap

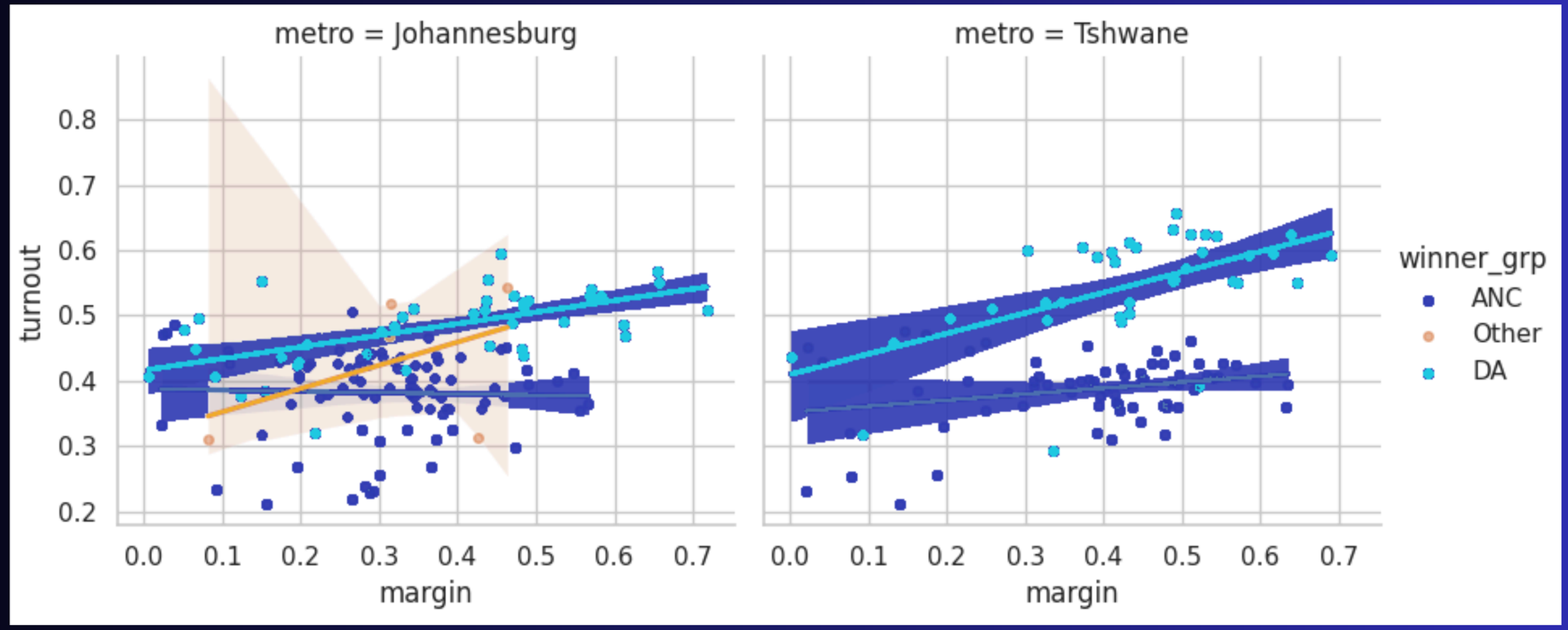
NOT YET TESTABLE AT WARD LEVEL

Youth share of adults (Census, metro-level, real): JHB 46.1%, Tshwane 43.7% re-export still returns District level, not Ward.

What would change it:

IEC 2026 registration by age, per ward.

Supporting evidence



Interaction model (M3): margin vs turnout, by winner group and metro. The slope only shows up in DA wards, matching the H2 verdict above.

08 · LIMITS & HONESTY

Named up front, and handled not hidden

Limitation	Effect	Mitigation
Ward-level youth turnout isn't published	Can't model youth turnout directly	Split into two explicit targets: overall turnout (T1) and youth registration (T2)
Census exported at municipality level	Only 2 data points — H1 untestable so far	Audit caught it early; ward-level re-export requested
Ward boundaries changed 2016→2021	Same ID ≠ same area	Voting-district crosswalk with printed coverage
Small sample (242 wards)	Wide intervals, overfitting risk	Regularised models, one-SE rule, VD-level checks (n=1,640)
Ecological inference	Ward patterns ≠ individual behaviour	Stated plainly in every conclusion

09 · DEPLOYMENT

From notebook to something IEC, journalists and voters can actually open

The pipeline already exports what the app needs: turnout_model.joblib, model_metadata.json, master_ward_table.csv, forecast_2026.csv

Ward explorer

JOURNALISTS, VOTERS

Turnout, winner, margin, ENP and deprivation, browsable by ward

Priority list

IEC, CIVIL SOCIETY

Ranks wards by low forecast turnout × low youth registration

Policy simulator

POLICYMAKERS

Adjust deprivation inputs, see the predicted turnout shift labelled as association, not cause

Four small decisions doing a lot of work

Tent-station proxy

IEC station names themselves reveal informal settlements 155 of 1,640 voting districts, no external data needed.

2016 to 2021 crosswalk

Re-aggregates old results onto new ward boundaries, instead of matching IDs that no longer mean the same place.

Defection vs disengagement

Separates a change in incumbent vote share from a change in turnout, to test H2's mechanism, not just its outcome.

One-SE + VIF rules, set in advance

Model choice was decided by pre-stated criteria — not picked after seeing which one looked best.

Thank You

Turning data into insights, and insights into better decisions.

Presented by:

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